Code No: 118BJ **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, July - 2023
EMBEDDED SYSTEMS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

Used paper July/Aug-2023

PART – A

(25 Marks)

1.a)	What is a timer of 8051?	[2]
b)	List the general-purpose registers available in 8051 architecture.	[3]
c)	What is the difference between conditional and unconditional jump?	[2]
d)	Add 84h to RAM locations 17h and 18h.	[3]
e)	What is a semaphore?	[2]
f)	Give the three rules to decide whether a function is reentrant.	[3]
g)	What is a hard time scheduling?	[2]
h)	List the goals of the typical testing process.	[3]
i)	What is internet security?	[2]
j)	Write the important features of SHARC.	[3]

PART – B

(50 Marks)

2.a)	Explain the challenges in embedded system design.	
b)	Write the interrupts of 8051.	[5+5]
	OR	
3.a)	Explain the different modes of timers.	
b)	Discuss about I/O ports of 8051.	[5+5]
4.	Explain the logical instructions of 8051 with examples.	[10]
	OR	
5.	Explain the Jump instructions of 8051 with examples.	[10]
6.	Discuss the interfacing of the keyboard with a neat diagram.	[10]
	OR	
7.	Explain two rules that must follow by the interrupt routines in most RTOS environments.	[10]
8.	Explain the encapsulation of semaphores and queues with the required code.	[10]
	OR	
9.	Discuss different ways to get embedded software into the target system.	[10]
10.	Compare the Instruction level parallelism of ARM and SHARC.	[10]
	OR	
11.	Discuss the operation and requirements of elevator controller design.	[10]

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